SANDAG · 7TH-CYCLE RHNA · METHODOLOGY DEVELOPMENT

An objective methodology, built in the open

Seven phases from a passing replication of the 6th-cycle allocation to a package HCD can review — leaning on true physical capacity constraints as far as the fair-housing objective permits, computed entirely from public data.

Milestone 0 **complete — replication passing** Planning period **Apr 2029 – Apr 2037**

Draft to HCD **late 2027**

The rule everything follows from

> **Capacity determines *where* units go, never *how many* a jurisdiction receives.**
>
> Three capacity measures are named in `Gov. Code §65584.04(e)` and are therefore not merely permitted but **required to be considered**: sewer and water limits arising from state or federal action; availability of land suitable for development and protected lands; and emergency evacuation route capacity, wildfire risk and sea level rise.
>
> Applied as a **reducer** of a jurisdiction's total, capacity collides with the fair-housing objective at `§65584(d)(5)`. Applied as a **siting input inside a fixed total**, it satisfies `(e)` and `(f)` without touching `(d)(5)`. Every phase below is built to hold that line.

What every phase must produce

A phase is not done until a person who has not read the code can check the result. Each ends in a markdown report and its backing CSVs, carrying:

- every number's source, vintage, and the URL it came from;
- a reconciliation to a figure the publishing agency states independently, wherever one exists;
- the SHA-256 of every input file actually used;
- a machine acceptance test that fails the build if the check breaks.

The **How a person checks it** note on each phase below is the point of this plan. If a phase cannot offer one, it is not ready to start.

The phases

7 PHASES · 2 GATES

1 AFFH backbone

4 WEEKS

Establish the fair-housing measuring stick before anything is measured against it. Ingest the TCAC/HCD Opportunity Map at tract level, implement the resource-only allocation as a permanent baseline, and build the AFFH gate — the pipeline refuses to emit any methodology whose share of lower-income units in High and Highest Resource tracts falls below it.

Output `reports/affh_baseline.md`

HOW A PERSON CHECKS IT

TCAC publishes its own San Diego County tract counts per resource category. Ours must match exactly, and the report shows both side by side.

2 Constraint inventory

4 WEEKS

Get every capacity, safety and health layer onto tracts: CAL FIRE hazard severity zones, FEMA floodway and 100-year flood, USGS sea level rise scenarios, protected lands, and CalEnviroScreen. Shares are of *residential* land, not raw area, using the block-level housing weights the crosswalk already computes.

Output `reports/constraint_inventory.md`

HOW A PERSON CHECKS IT

CAL FIRE publishes hazard acreage by county and CPAD publishes protected acreage. The report reconciles our totals to theirs and shows the difference.

KNOWN WEAK POINT, FLAGGED NOW

The sewer and water measure is the thinnest of the three statutory capacity factors. The statute counts only limits from state or federal action, and there is no clean open dataset of those. **Do not let the weight structure assume this measure lands.**

3 The three-axis correlation study

CRITICAL PATH 3 WEEKS

Answer, with evidence, the question that determines the entire methodology: **in San Diego, does physical capacity run with or against opportunity?** Cross-tabulate every capacity, safety and health measure against resource category, weighted by housing units.

The hypothesis: wildfire and evacuation constraint concentrate in affluent foothill and exurban tracts, and sea level rise on expensive coast — so a capacity-as-reducer methodology would move units *out of* high-resource areas and *into* the lower-resource, more segregated urban core.

Safety and health are reported separately and must not be combined. Health burden is expected to run the opposite way from wildfire. If so, AFFH-compliant siting improves health outcomes while worsening wildfire exposure — a more credible and more useful finding than either alone, which merging them into one index would destroy.

Output `reports/capacity_opportunity_correlation.md`

HOW A PERSON CHECKS IT

Every cell is a cross-tab of two published CSVs from Phases 1 and 2 — recomputable in a spreadsheet. No modelling, no weighting, no parameters. This phase deliberately contains no judgement calls, so its results cannot be argued with on methodology grounds.

GATE A · CHAIR DECISION

Is the orthogonal design a preference, or a requirement?

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This report goes to the chair before any design commitment. It determines whether capacity can carry weight in a jurisdiction's total at all, and gives evidence to show a mayor who asks why it cannot.

4

Street network and evacuation capacity `8 WEEKS`

Road network from OpenStreetMap, retaining functional class, lane count and intersection geometry. From it: a link and intersection capacity index per tract, a per-tract evacuation capacity and clearance-time estimate, and an explicit, separable calibration step against published traffic counts.

The Highway Capacity Manual is a licensed TRB publication and is **not used**. The parameters it is known for are reproduced in free federal documents, and each must cite the public source: `FHWA HPMS Field Manual, Appendix N` for capacity parameters, and `NRC NUREG/CR-7002 Rev. 1` for the evacuation-time methodology. This is the same discipline applied to methods rather than data — an HCD reviewer can open every source.

Output `reports/network_capacity.md` `reports/evacuation_capacity.md`

HOW A PERSON CHECKS IT

The calibration report lists every public traffic count station in the county with the modeled volume beside the observed count, and the residual. A traffic engineer can audit it station by station. Uncalibrated tracts are labelled uncalibrated rather than quietly carrying a default.

5

Jobs adjustments `5 WEEKS · PARALLEL`

The three adjustments specified but not yet written, each toggleable and each logged: multi-site employer redistribution, QCEW reconciliation to county sector totals, and seasonal annualisation to full-time equivalents.

Output `reports/jobs_adjustments.md`

HOW A PERSON CHECKS IT — A KNOWN-ANSWER TEST AGAINST REAL HISTORY

The 6th-cycle appeals process found exactly one error: two Navy installations wrongly treated as remote stations of Naval Base San Diego, and North Island's jobs needing an 80.5/19.5 split. It took a phone call to Naval Facilities Engineering jobs needing an 80.5/19.5 split. It took a phone call to Naval Facilities Engineering Command, surfaced four months after the draft allocation issued, and moved 135 units.

The acceptance test is whether the multi-site detector finds that same error independently, from open data, with no knowledge of the answer. If it passes, the case for computing this in the open is made in one sentence to any board member.

6 The allocator, six income categories, and the frontier `6 WEEKS`

Turn the parameter files into mechanism. Build the general tract-scored allocator; add the six income categories with acutely low and extremely low held within 3% of proportionality to very low; then sweep the AFFH gate from unconstrained to maximum, solving at each level for the allocation that minimises hazard exposure.

Output `reports/frontier.md`

HOW A PERSON CHECKS IT

Every point on the curve is a committed parameter file. Rerunning any one reproduces its point byte for byte. The slope at the chosen point is the exchange rate between fair-housing performance and hazard exposure, in units, at the margin — the chair's question answered as a rate rather than a verdict. Where the frontier is flat there is no trade-off at all, and the report says so plainly.

GATE B · BOARD DECISION

The board picks a point on the frontier

Every point on the curve is lawful. The choice among them is political, and it should be made in the open with the numbers visible.

7 The HCD package `3 WEEKS`

The methodology appendix, generated from the code's own docstrings rather than written: every factor, its source URL, vintage and rule, in the format `§65584.04(f)` requires — plus how each `§65584.04(e)` factor was incorporated and how the methodology furthers each `§65584(d)` objective.

Output `reports/methodology_appendix.md`

HOW A PERSON CHECKS IT

Every factor traces to a source URL, a vintage, and a checksum. A reviewer can download any input and confirm the hash matches the one the report was generated against.

Sequence

ONE ANALYST FROM SEP 2026

PHASE	FINISHES	NOTE
1 · AFFH backbone	Oct 2026	
2 · Constraint inventory	Nov 2026	
3 · Correlation study	Dec 2026	Critical path
Gate A	Dec 2026	**Chair decides the design**
5 · Jobs adjustments	Feb 2027	Parallel with 4
4 · Network and evacuation	Mar 2027	Largest phase
6 · Allocator and frontier	May 2027	
Gate B	May 2027	**Board picks a point**
7 · HCD package	Jul 2027	

That leaves roughly a quarter of slack before a draft methodology needs to reach HCD in late 2027. The slack is deliberate: `§65584.04(i)` gives HCD 60 days to review and, if it finds the methodology does not further the objectives, a further 45-day revision cycle. **Phase 3 is only three weeks of work, but everything downstream is shaped by its result.** If schedule pressure appears, protect its inputs first.

Recommended alongside the technical work

Talk to HCD informally before anything is public

Hearing "we would have concerns with that" in a staff call costs nothing. Hearing it in the `§65584.04(i)` findings letter costs a revision cycle. The concept to test is the orthogonal design: capacity as a siting input inside a fixed total.

Have counsel check one citation

The statutory guardrails cite `§65584.04(e)(2)(B)` for the three prohibited justifications, taken from SANDAG's own restatement in its adopted 6th-cycle methodology. The section has been amended repeatedly and that subdivision letter may have moved. The substance is right; the pincite will appear in the appendix HCD reads.

Expect the framing fight

A chart showing hazard exposure rising with fair-housing performance will be quoted without its context. The defences are built into the plan: report both directions of the trade-off, keep safety and health on separate axes, publish the mitigation-adjusted result alongside the raw one, and never emit a single scalar "cost of AFFH". The strongest position — and the true one — is that this reconciles competing factors the statute itself imposes, with the arithmetic open.

Open questions that block work

01 **Six-category control totals.** HCD has not issued a 7th-cycle determination. Until it does, does the pipeline use a placeholder RHND, or the 6th-cycle determination resplit into six categories?

`Blocks Phase 6`

02 **Wildfire mitigation assumption.** A unit built to current WUI standards is not the risk the surrounding older stock is. Does the safety axis report raw exposure, mitigation-adjusted exposure, or both? Recommend both, raw as the headline.

`Shapes Phases 3 and 4`

03 **AMI-based affordability.** The jobs-housing fit metric uses wage bands fixed in nominal dollars, drifting against area median income yearly. Keying it to HCD's State Income Limits would answer the statutory question directly — which vintage gets pinned?

`Shapes Phase 5`

04 **Transit factor.** 65% of the 6th-cycle allocation ran on proprietary model output counting *stations*. Replicate the station count from open transit feeds (reproducible, keeps a crude measure), or replace it with a jobs-reachable-in-45-minutes accessibility surface (better measure, not comparable to the 6th cycle)?

Blocks Phase 6

Milestone 0 replication: max error 2 units across 76 cells of 171,685

All inputs public · all runs byte-reproducible